

Mop a Floor Zone SOP (1x Zone, in situ)

One episode mops one floor zone, in place.

The base is **passive**. It has no drive of its own. It is pushed by hand up to the open front of the zone and locked there. Nothing is carried away to a table. Everything the episode touches is already on the floor when recording starts.

The zone is worked **as found**. The floor is dirty from edge to edge. The four things the episode uses are the bucket, the water jug, the mop, and the caution sign. They all sit on the floor around the open front of the zone. The episode ends with the bucket filled, the floor mopped in three S laps and one edge pass, the mop back in its rest, and the caution sign standing open at the front of the zone.

This is an in-situ task. Four rules follow from that.

First, the zone is a fixed bay in a run of furniture. So **every approach is from the front, level**. The arms come in over the open front of the bay and go back out the same way. Nothing reaches in over the cabinets and nothing swings down from above.

Second, **the cabinets, the kick rail, and the plinths are never loaded**. No gripper, wrist, or forearm leans on them. The mop head is never pushed up onto them.

Third, **the mop head stays on the floor**. It is lifted only over the bucket, at the dip and the wring, and once at the end into its rest.

Fourth, **nothing is carried across the zone**. The bucket, the jug, and the sign are all worked where they stand.

The order never changes. Fill the bucket. Dip and wring the mop. Run three S laps. Run one edge pass. Rest the mop. Stage the sign. The mop is dipped and wrung again before the edge pass, with no exception.

The two arms never cross. The bucket stands directly in front of the base. The **left gripper works its left side**. The **right gripper works its right side**. The left gripper always stays left of the right gripper.

Arm roles are fixed and never swap. The **right gripper is the tool gripper**. It carries and pours the jug. It holds the mop handle for the dip, the wring, all three laps, and the edge pass. It stands the mop back in its rest. The **left gripper is the steady gripper**. It holds the bucket still by its left side lug for the pour and for every wring. It opens the caution sign. It never touches the mop or the jug.

The laps run **from the left of the bay rightward**, so each new lap is laid beside the one just done and the head never travels back over floor it has already mopped.

The right gripper holds the mop at the **grip band**, up the handle, so the head reaches the back of the bay while the arm stays in its own box in front of the base. **The handle does the reaching, not the arm.**

Setup

Complete the base positioning and both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Base positioning

The base is **passive**. It has no drive of its own. It is pushed into place by hand. It is never steered, nudged, or repositioned once recording starts. It is parked once, before recording, and does not move again until the episode is over.

1. Push the base by hand up to the open front of the bay. Stop it **square to the kick rail**, so the kick rail runs straight across the frame and neither plinth sits nearer than the other.
2. Stop it **centered on the bay**, so the left plinth and the right plinth are the same distance out from the middle of the base.
3. Stop it **close enough** that the right gripper puts the mop head at the kick rail without the arm extending. Stop it **far enough** that no arm, wrist, forearm, or part of the base touches the plinths, the kick rail, the cabinets, the bucket, or the sign when both arms work at once.
4. Check the **zone reach**. With the base parked, the **right gripper** holds the mop at the grip band and puts the head flat on the floor at the back-left corner of the bay, at the back-right corner, and at both front corners. It runs one full lap from the front edge to the kick rail. It does all of this without extending and without the handle touching the cabinets above.
5. Check the **bucket reach**. With the base parked, the **left gripper** reaches the bucket's **left side lug** and the **right gripper** reaches the **wringer on the bucket's right side**, both without extending, and without either arm reaching past the other.
6. Check the **jug reach**. With the base parked, the **right gripper** lifts the jug off the jug spot and holds it over the bucket without extending.
7. Check the **sign reach**. With the base parked, the **left gripper** reaches the grab hole of the folded sign and turns the top panel fully open without extending and without reaching past the middle of the base.
8. Lock or brake the base. Push it firmly once by hand. It must not roll, creep, or turn.
9. If any of lines 1 to 7 fails, push the base to a new park by hand and start again at line 1. If line 4 still fails after a re-park, the bay is too wide or too deep to work from one park. Do not work it.

The base stays locked and still for the whole episode. Nothing moves it. No arm leans on the furniture hard enough to shift it. Nothing touches it by hand after recording starts. It is never repositioned mid-task. A base that moves after recording starts ends the episode.

Hardware checklist

1. Cameras are on and recording.
2. The environment camera is centered on the bay. Its frame includes the whole floor of the bay from the left plinth to the right plinth and from the kick rail to the front edge, the kick rail itself, both plinth ends, the bucket with its wringer, the jug spot, the mop rest, and the folded caution sign.
3. The camera reads the floor from the front at an angle where dirt, dust, scuff marks, wet floor, and dry floor are all visible. If a glare patch on the floor hides part of the bay, move the camera until the whole bay reads.
4. The camera reads the line where the floor meets the kick rail and where it meets each plinth. You can see whether the edge pass ran tight to the edge.
5. The camera reads the mop head from the front. You can see whether the pad is flat on the floor and whether it is dripping.
6. The camera reads the fill line inside the bucket.
7. Both arms are at home with grippers open.
8. The cabinets, the kick rail, and the plinths are fixed firm. They do not move or rock when the mop head is pushed against them.
9. The **right arm** reaches the jug spot, the right side of the bucket, the wringer, the mop rest, and the whole floor of the bay from the kick rail to the front edge and from the left plinth to the right plinth. It reaches all of them without extending to a joint limit.
10. The **left arm** reaches the bucket's left side lug and the folded sign at the front-left of the bay. It reaches both without extending to a joint limit. It never needs to reach the mop rest, the jug spot, or the wringer.
11. Each arm comes at the bay **from the front, level**: in over the open front and back out the same way. No part of the arm passes over the cabinets or touches the kick rail, a plinth, or a cabinet door.
12. Both arms work the bucket at the same time without touching each other, the bucket, or a plinth, with the left gripper on the bucket's left and the right gripper on its right.
13. If a place cannot be reached, re-park the base by the Base positioning steps until lines 9 to 12 hold.

Materials checklist

1. The **bay** is one open floor space in a run of fixed furniture. The **kick rail** closes it at the back. A **plinth end** closes it at the left and another at the right. The front is open. It is the only bay in frame.

2. The bay is narrow enough that **three laps of the mop head with overlap cover it from the left plinth to the right plinth**. It is shallow enough that the right gripper runs the head from the front edge to the kick rail in one move without extending.
3. The floor of the bay is **hard, sealed, and flat**. There is no rug, no mat, no drain, no grating, and no loose tile that rocks under the head.
4. The floor is **dirty from edge to edge and from the kick rail to the front edge**. Visible dust, scuff marks, and dry footprints cover the whole bay. The dirt reads on camera from the front.
5. The floor is **dry** at episode start. There is no spill, no standing water, and no wet patch anywhere in the bay.
6. The floor carries no grit, crumbs, wrappers, or loose debris. Nothing a mop would push into a pile or skid on is in the bay.
7. The floor has no chip, no stain in the floor itself, and no worn patch that a correct lap will not lift.
8. Nothing stands, hangs, or leans inside the bay. No cabinet door or drawer is open into it.
9. The **bucket** is a rectangular mop bucket. It stands on the floor **directly in front of the base, just outside the front edge of the bay**, centered on the base and clear of the bay.
10. The bucket carries a moulded **side lug on its left side**, facing the left arm, and a **wringer basket clipped on its right side**, facing the right arm. Its base is wide enough that it does not tip when the head is pressed into the wringer.
11. The bucket is **empty and dry** at episode start. It carries a **fill line** moulded inside it that reads on camera from the front.
12. The left side lug sits inside one gripper's opening, so the **left gripper** closes on it and holds the bucket still without the bucket tipping or turning in the grip.
13. The **water jug** is a rigid open-topped jug with a pouring lip and a rigid handle. It stands upright on the floor on its **jug spot**, on the floor **to the right of the bucket**, outside the bay.
14. The jug holds enough water to bring the bucket to its fill line in **one pour**, and no more than the bucket takes.
15. The jug handle sits inside one gripper's opening, so the **right gripper** closes on it and the jug hangs upright without sagging or turning in the grip. The jug does not tip when a gripper closes on it. One gripper carries it on its own without help.
16. The **mop** is a flat mop: a rectangular **head** on a rigid **handle**, with a clean, dry **pad** already fitted. The head is hinged so it lies flat on the floor and follows the handle. It does not spin loose in the hinge.
17. The head is wide enough that three laps with overlap cover the bay from plinth to plinth. It is narrow enough to fit inside the bay with clear space at each side.
18. The handle carries a moulded **grip band**. The band sits inside one gripper's opening. The handle does not turn or slip in the grip when the head is pushed along the floor.

19. The **mop rest** is a fixed cradle on the outside of the **right plinth**, clear of the bay. The mop stands in it at episode start and at episode end, handle in the cradle, head flat on the floor.
20. The pad is absorbent enough to hold water for three laps and one edge pass without the head going dry. It does not shed fibres onto the floor.
21. The **caution sign** is a two-panel folding A-frame with a hinge along its top edge, feet along the bottom edge of each panel, and a **grab hole** in the middle of one panel.
22. The sign lies **folded flat on the floor at the front-left corner of the bay, just outside it**, with the hinge toward the kick rail and the printed face of the top panel facing up.
23. The grab hole sits inside one gripper's opening, so the **left gripper** closes on it and lifts the top panel on its own.
24. The bottom panel has **rubber feet that grip the floor**, so the sign opens under one gripper without the bottom panel sliding, and the second arm is never needed.
25. The sign stands on its own when it is opened. It does not fold back. It is light enough for one gripper to lift its top panel.
26. The floor under the base and around the bucket is level, dry, and clear.

Workspace layout

Nothing anywhere is marked or taped out. You judge every place below by eye against the furniture itself: the kick rail, the two plinths, and the tools already in place.

- **Bay:** the open floor space being mopped. Also called the **zone**. The kick rail bounds it at the back, a plinth end at the left and at the right, and it is open at the front.
- **Kick rail:** the fixed board that closes the back of the bay. It is never leaned on and never pushed. The mop head never rides up onto it.
- **Plinth:** the fixed end board that closes the left or the right of the bay. It is never leaned on and never pushed.
- **Cabinets:** the run of furniture above and behind the bay. Nothing reaches over them and nothing touches them.
- **Back edge, left edge, right edge:** the three lines where the floor meets the kick rail, the left plinth, and the right plinth.
- **Front edge:** the open line across the front of the bay, level with the front of the base. It has no board along it.
- **Corners:** the four places where two edges of the bay meet: back-left, back-right, front-left, front-right.
- **Bucket:** the mop bucket, standing on the floor directly in front of the base, just outside the front edge of the bay. Its **left side lug** faces the left arm. Its **wringer** is clipped on its right side and faces the right arm. Its **fill line** is moulded inside it and the pour is judged against it.

- **Jug spot:** the place on the floor to the right of the bucket where the water jug stands at episode start and at episode end. **Right gripper only.**
- **Mop rest:** the cradle on the outside of the right plinth where the mop stands at episode start and at episode end. **Right gripper only.**
- **Sign place:** the floor just outside the front-left corner of the bay, where the sign lies folded at episode start and stands open at episode end. The sign is never carried away from it. **Left gripper only.**

Arm lanes

The two arms never cross. This holds for the whole episode.

- The **left gripper works the left:** the bucket's left side lug and the sign at the front-left. The **right gripper works the right:** the jug, the wringer, the mop, and the mop rest.
- The **left gripper always stays left of the right gripper.** Neither arm reaches over, under, around, or past the other. Neither arm reaches across the front of the other arm's body.
- The **left arm never enters the bay.** It stays outside the front edge for the whole episode.
- The **right gripper stays in front of the base** while it mops. The **handle** carries the head out to the left of the bay, not the arm. The right arm never swings out past the middle of the base.
- Nothing is handed from gripper to gripper. Only one arm moves an object at a time.

Arm assignments

- **Right gripper. The tool gripper.** Lifts the jug and pours it. Puts the jug back. Takes the mop from its rest. Dips the head. Presses the wring. Runs all three laps and the edge pass. Stands the mop back in its rest. It never holds the bucket and it never touches the sign.
- **Left gripper. The steady gripper.** Holds the bucket still by its left side lug for the whole pour and for every wring. Opens the caution sign at the end. It never touches the mop, the jug, the wringer, or the floor of the bay.

Vocabulary

- **Front approach:** the gripper comes in over the open front of the bay, level, and goes back out the same way. It never reaches in over the cabinets and never swings down from above. Every action in the bay is a front approach.
- **Head:** the flat rectangular part of the mop that touches the floor. **Pad** is the cloth face on the bottom of it. **Handle** is the rigid pole above it. **Grip band** is the moulded band on the handle where the right gripper closes. A gripper closes on the grip band only.
- **Head flat:** the whole pad touches the floor at once. No corner is lifted and no edge rides up.
- **Lap:** one run of the head along the bay from the front edge straight back to the kick rail, or from the kick rail straight out to the front edge. The head stays flat the whole way, in one

steady push.

- **Steady push:** the head moves at one speed. It does not stop, jerk, speed up, lift, or turn.
- **Step over:** at the end of a lap, the head is slid sideways to the start of the next lap without leaving the floor. It is a slide, not a lift.
- **S laps:** the three laps and the two step-overs, run as one unbroken shape: back, over, out, over, back. The head does not leave the floor from the moment it enters the bay to the moment the third lap ends.
- **Overlap:** the new lap is set so that **about a third of the head sits on the wet strip** the last lap left. The rest sits on dry floor. You judge it by eye against the line between wet and dry.
- **Wet strip:** the band of floor a finished lap has left behind. It reads evenly wet from the front, against the dry floor beside it.
- **Dry band:** a stretch of floor between two wet strips, or at an edge, that is still dry. It is what the lap check looks for.
- **Clear:** no dry band, no dust line, no scuff mark, and no streak of dirty water is visible on that stretch of floor when it is read from the front. This is what a finished lap and a finished edge both look like.
- **Edge pass:** one unbroken run of the head round all four edges of the bay, with the long side of the head kept tight against the edge it is on.
- **Tight against the edge:** the long side of the head touches the line where the floor meets the kick rail or a plinth, with no dry band left between the head and that line, and without the head riding up onto the board.
- **Dip:** the head is lowered into the water in the bucket and moved up and down twice, so the whole pad is wet.
- **Wring:** the head is set in the wringer and the handle is pressed down twice, so the water is squeezed out of the pad.
- **Wrung:** the pad is damp and no water runs off the head when it is held clear of the bucket for 2 seconds.
- **Fill line:** the line moulded inside the bucket that the pour is judged against.
- **Panels:** the two boards of the caution sign. The **bottom panel** stays flat on the floor while the sign is opened. The **top panel** is the one lifted. **Grab hole** is the hole in the top panel the left gripper closes on.
- **Sign standing open:** the two panels stand as an A on their own feet. The sign does not fold back when the gripper lets go. The printed face points out to the front.
- **Loaded:** pressed, leaned on, or pushed hard enough to move. The cabinets, the kick rail, and the plinths are never loaded.

Steps

Step 1: Fill the bucket

Goal: the bucket holds water to its fill line. The jug stands back on its spot. No water is on the floor.

This is the one step where both grippers work at once. The **left gripper holds the bucket still from its left side**. The **right gripper pours from its right side**. Nothing enters the bay.

- With the **left gripper**, come in from the front and close on the **left side lug** of the bucket. Hold the bucket still on the floor. Do not lift it, do not tip it, and do not drag it.
- With the **right gripper**, come in from the front and close on the **jug handle**.
- Lift the jug straight up off the jug spot and keep it **upright** while it moves. Do not carry it over the bay, over the mop rest, or over the sign. Do not carry it over the left arm.
- Bring it over the bucket, clear of the wringer, with the pouring lip inside the rim of the bucket.
- Tip the jug slowly and pour in one go until the water reaches the **fill line**. Keep the lip inside the rim the whole time.
- Bring the jug back upright. Hold it over the bucket for 2 seconds so the last drips fall inside. Then take it straight back to the **jug spot**, stand it upright, open the **right gripper**, and draw the arm out to the front.
- Open the **left gripper** off the side lug and draw the arm out to the front.

Check: the water is at the fill line. The jug stands upright on its spot. The bucket has not moved off its place. The floor around the bucket is dry.

If the water is short of the fill line, take the jug up again with the **right gripper**, hold the bucket with the **left gripper**, and pour again until it reaches the line. If water has gone on the floor outside the bay, end the episode and reset.

Expected state: bucket filled to the fill line and standing where it started. Jug upright on its spot. Mop in its rest. Sign folded flat at the sign place. Bay dry and dirty from edge to edge.

Step 2: Dip and wring the mop

Goal: the **right gripper** holds the mop at the grip band with the pad wrung damp, and the head is on the floor.

2.1 Take up the mop

- With the **right gripper**, come in from the front and close on the handle at the **grip band**. Do not touch the head and do not touch the pad.
- Lift the handle out of the mop rest and keep the handle **upright**. Do not swing it over the bucket, over the cabinets, or across the bay.

2.2 Dip the head

- With the **left gripper**, come in from the front and close on the **left side lug** of the bucket. Hold the bucket still for the whole of 2.2 and 2.3.
- With the **right gripper**, lift the head just clear of the floor and lower it straight down into the water in the bucket, clear of the wringer.
- Move it **up and down twice** in the water, so the whole pad is wet. Do not stir it round and do not knock it against the sides of the bucket.

2.3 Wring the head

- With the **right gripper**, lift the head straight up out of the water and set it into the **wringer** on the bucket's right side.
- Press the handle straight down **twice** to squeeze the pad. Press only as hard as the wringer takes. Do not lever the handle sideways and do not push the bucket over.
- Lift the head straight up out of the wringer and hold it over the bucket for **2 seconds**.
- Lower the head straight down onto the floor **just outside the front edge of the bay, in front of the base**. Then open the **left gripper** off the side lug and draw that arm out to the front.

Check: no water runs off the head in those 2 seconds, and the pad is damp, not dripping. If water is still running off, set the head back in the wringer with the **right gripper** and press the handle down twice again.

Expected state: mop in the right gripper with a wrung damp pad. Head flat on the floor just outside the front edge. Bucket filled and standing where it started. Left arm clear. Bay still dry and dirty.

Step 3: Mop the zone in three S laps

Goal: the whole bay is mopped in three overlapping laps, left to right, run as one unbroken S.

Run 3.2 and 3.3 in turn: **lap 1, step over, lap 2, step over, lap 3**. There is no step over after lap 3.

The head does not leave the floor at any point in Step 3. The left arm stays at home, clear of the bay, for the whole of it. The right gripper stays in front of the base, and the handle carries the head out to the left, not the arm.

3.1 Bring the mop into the bay

- With the **right gripper**, slide the head along the floor from in front of the base, across the **front edge** of the bay, to the **front-left corner**. Keep the pad flat on the floor the whole way. Let the handle reach. Do not swing the arm out to the left.
- Set the head there with its long side across the bay and its left side **just clear of the left plinth**, not touching it.

Check: the head sits flat on the floor at the front-left corner, just clear of the left plinth, with the whole pad down and the handle held at the grip band. If a corner of the head is lifted, or the head is against the plinth, set it again with the **right gripper**.

3.2 Run one lap

- With the **right gripper**, push the head **straight back from the front edge to the kick rail** in one **steady push**, keeping the pad flat on the floor the whole way. This is how **lap 1** and **lap 3** run.
- On **lap 2**, pull the head **straight out from the kick rail to the front edge** in the same way.
- Stop the head **just short of the kick rail** on the way back, and level with the **front edge** on the way out. Do not push the head up onto the kick rail. Do not run it out past the front edge.
- Do not lift the head. Do not stop mid-lap. Do not jerk it. Do not turn it as it runs.
- Let the handle do the reaching. Do not extend the arm to follow the head to the kick rail.

Check: the strip the head has just laid reads evenly wet from the front, with no dry band left in it and no scuff still showing. If a dry band or a scuff shows, run that same lap again with the **right gripper**, once.

Expected state: one more wet strip on the floor. Head flat on the floor at the end of that lap. Left arm at home.

3.3 Step over to the next lap

- With the **right gripper**, slide the head sideways **to the right**, keeping the pad on the floor. Do not lift it.
- Set it so about a **third of the head sits on the wet strip** the last lap left and the rest sits on dry floor.
- On the step over into **lap 3**, also set the head so its **right side is just clear of the right plinth**, not touching it.

Check: the head sits flat, with about a third of it on the wet strip and the rest on dry floor. If it is set clear of the wet strip, or so far onto it that almost no dry floor is under it, slide it again with the **right gripper**.

Expected state after the third lap: the whole bay reads clear from the left plinth to the right plinth and from the kick rail to the front edge, apart from a thin dry band round the three edges. The head is flat on the floor at the back-right corner.

Check after the third lap: read the whole bay from the front. If a dry band shows between any two laps, slide the head onto it with the **right gripper** and run one more lap over that band, once, by Steps 3.2 and 3.3.

Step 4: Run the edge pass

Goal: all four edges of the bay are mopped tight to the line, in one unbroken run.

4.1 Dip and wring again

- With the **right gripper**, slide the head along the floor out of the bay, over the front edge, to the bucket in front of the base. Then dip and wring the mop again by Steps 2.2 and 2.3, with the **left gripper** holding the left side lug.

Check: no water runs off the head in those 2 seconds. If water is still running off, press the handle down in the wringer twice again with the **right gripper**.

4.2 Run the edge pass

- With the **right gripper**, slide the head back onto the floor at the **front-right corner** of the bay. Turn it so its long side lies **along the front edge**.
- Run the head round the bay in one unbroken move, keeping the pad flat and the long side **tight against the edge** it is on, in this order:
 - along the **front edge**, from the front-right corner to the front-left corner;
 - up the **left edge**, from the front-left corner to the back-left corner;
 - along the **back edge**, from the back-left corner to the back-right corner;
 - down the **right edge**, from the back-right corner to the front-right corner.
- At each corner, **turn the head by moving the handle round it**. Do not twist the wrist to turn the head. Do not lift the head off the floor to reset it.
- Keep the head **just off the boards**. The pad runs against the line where the floor meets the kick rail or the plinth. It never rides up onto the board itself.
- Do not lean the handle on the kick rail, a plinth, or a cabinet. Do not push hard enough to move them.
- Keep the right arm in front of the base for the whole pass. The handle reaches the left edge, not the arm.

Check: all four edges read **clear** from the front, with no dry band left between the wet floor and the kick rail or either plinth, and no water pushed up onto a board. If a dry band shows along an edge, run the head along that edge again with the **right gripper**, once.

Expected state: the whole bay reads clear from edge to edge and corner to corner. The kick rail and both plinths are dry and unmarked. The head is flat on the floor at the front-right corner. The bucket, the jug, and the sign are all where they started.

Step 5: Rest the mop

Goal: the mop stands back in its rest, outside the bay.

- With the **right gripper**, slide the head along the floor out of the bay, past the right plinth, to the **mop rest**.
- Lift the handle upright and stand it in the **cradle**, with the head flat on the floor beside the plinth.

- Open the **right gripper** and draw the arm out to the front. Do not lay the mop on the floor of the bay. Do not lean it on a plinth or a cabinet. Do not stand the head on the bay side of the right plinth.

Check: the mop stands in its cradle with the head flat on the floor outside the bay, and the **right gripper** is clear of it. If the mop leans, slips, or falls, take the handle at the grip band with the **right gripper** and stand it again.

Expected state: mop in its rest. Bucket filled and in place. Jug upright on its spot. Bay mopped clear. Sign still folded flat at the sign place.

Step 6: Stage the caution sign

Goal: the caution sign stands open at the sign place with its printed face out to the front.

The sign sits at the front-left of the bay. So the **left gripper** does this whole step on its own. The right arm stays at home. The sign is opened where it lies. **It is never lifted off the floor and never carried.**

- With the **left gripper**, come in from the front and close on the **grab hole of the top panel**.
- Lift the top panel up and over **toward the front**, turning it on the hinge, until the two panels stand as an A on their own feet. The bottom panel stays flat on the floor on its rubber feet.
- Hold the top panel until the sign stands on its own. Then open the **left gripper** and draw the arm out to the front.
- Do not slide the sign along the floor. Do not lift it off the floor. Do not set any part of it inside the bay.

Check: the sign stands open on its own feet at the sign place. It does not fold back when the gripper is clear. Its printed face points out to the front. If it folds back, tips, or stands on only one set of feet, close on the grab hole with the **left gripper** and set it again, once.

Expected state: sign standing open at the front-left of the bay. Both arms clear of it. Bay mopped clear. Mop in its rest. Bucket and jug in place.

Step 7: End the episode

- Confirm the floor. The whole bay reads clear from the left plinth to the right plinth and from the kick rail to the front edge, with no dry band, no scuff, and no streak of dirty water left on it.
- Confirm the edges. All four read clear, with no dry band left along the kick rail or either plinth.
- Confirm the furniture. The kick rail, both plinths, and the cabinets are dry, unmarked, and stand as they did at the start.
- Confirm the tools. The mop stands in its rest with the head flat on the floor. The bucket stands filled where it started. The jug stands upright on its spot.

- Confirm the sign. It stands open on its own feet at the sign place, printed face out to the front, clear of the bay.
- Confirm the base has not moved.
- Correct any failed check before ending.
- Return both arms home with grippers open. Then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Empty the bucket, rinse it, dry it inside and out, and stand it back in front of the base just outside the front edge of the bay, left side lug facing the left arm and wringer clipped on its right side.
2. Refill the water jug with enough water to bring the bucket to its fill line in one pour. Stand it upright on the jug spot, to the right of the bucket.
3. Fit a clean, dry pad on the mop head and take the used one away. Stand the mop in its cradle with the head flat on the floor.
4. Dry the floor of the bay fully. Then put fresh dirt on it: dust, scuff marks, and dry footprints from the left plinth to the right plinth and from the kick rail to the front edge. Check it reads on camera from the front.
5. Dry the kick rail, both plinths, and the floor around the bucket. Wipe up any water outside the bay.
6. Fold the caution sign flat and lay it back at the sign place, hinge toward the kick rail, printed face up.
7. Replace a pad that is torn, shedding, or that no longer holds water for three laps and an edge pass.
8. Replace a mop whose head no longer lies flat on the floor, whose hinge spins loose, or whose handle turns in the grip.
9. Replace a bucket that tips when the head is pressed in the wringer, whose left side lug no longer sits inside one gripper's opening, or whose wringer no longer squeezes the pad damp in two presses.
10. Replace a jug that leaks, that tips when gripped, or whose handle no longer sits inside one gripper's opening.
11. Replace a sign that folds back on its own, that will not stand on its own feet, whose grab hole no longer sits inside one gripper's opening, or whose bottom panel slides on the floor when the top panel is turned up.
12. Check the kick rail, the plinths, and the cabinets are still fixed firm and do not rock when the head is pushed against them.
13. Check the floor of the bay is clear of grit, crumbs, and loose debris.

14. Check the base is still locked and parked square. Then run the Base positioning steps and both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Base moved during the episode

- **Visible cue:** the bay shifts in frame, the kick rail changes angle or size in frame, or the base rolls, creeps, or turns at any point after recording starts.
- **SOP rule broken:** Steps 1 to 7. The base is parked and locked before recording and stays still for the whole episode.
- **Coaching note:** park it, lock it, push-test it, then start recording. A base that drifts is a lost episode.

Violation: Arms crossed

- **Visible cue:** the **left gripper** goes to the right of the right gripper, touches the jug, the wringer, the mop, or the mop rest, or enters the bay. The **right gripper** goes to the left of the left gripper, touches the bucket's left side lug, or touches the sign. One arm reaches over, under, around, or past the other, or across the front of the other arm's body. The right arm swings out past the middle of the base to follow the mop head.
- **SOP rule broken:** Arm lanes and Arm assignments. The left gripper works the left, the right gripper works the right, and the left gripper always stays left of the right gripper.
- **Coaching note:** each arm stays in its own lane. The mop handle reaches for you, so the arm never has to.

Violation: Not a front approach

- **Visible cue:** an arm reaches into the bay over the cabinets or down from above instead of in over the open front. Or the mop handle is swung up over the cabinets or across the bay instead of the head being slid along the floor.

- **SOP rule broken:** Steps 1 to 6. Every action in the bay is a front approach: in over the open front, level, and back out the same way.
- **Coaching note:** in over the front, do the work, out over the front. Nothing goes in from above.

Violation: Furniture loaded

- **Visible cue:** the kick rail, a plinth, or a cabinet moves, rocks, or shifts. A gripper, wrist, or forearm leans or rests on one of them. The mop handle is leaned against one. Or the head is pushed hard enough to move a board.
- **SOP rule broken:** Steps 3 to 6. The cabinets, the kick rail, and the plinths are never loaded, and the head is pushed only hard enough to keep the pad flat.
- **Coaching note:** the furniture holds nothing up for you. Light pressure, and let the arm carry its own weight.

Violation: Head lifted off the floor inside the bay

- **Visible cue:** the head comes off the floor at any point between entering the bay in Step 3.1 and leaving it in Step 5. It is lifted over a wet strip, lifted to reset a lap, lifted to turn at a corner, or hopped across the bay.
- **SOP rule broken:** Steps 3.1 to 5. The head stays on the floor from the moment it enters the bay to the moment it leaves it. It is lifted only over the bucket at the dip and the wring.
- **Coaching note:** slide, never lift. A head that hops leaves a gap you cannot see until it dries.

Violation: Wrong order

- **Visible cue:** the mop enters the bay before the bucket is filled. A lap is run on a dry or unwrung head. The edge pass is run before the third lap. The mop is rested before the edge pass. Or the sign is staged before the mop is in its rest.
- **SOP rule broken:** Steps 1 to 6. Fill the bucket, dip and wring, run three laps, run the edge pass, rest the mop, then stage the sign.
- **Coaching note:** fill, wring, laps, edges, rest, sign. Nothing runs dry and nothing is staged early.

Violation: Bucket filled wrong

- **Visible cue:** the pour stops short of the fill line or runs past it. The **right gripper** tips the jug with the lip outside the rim. Water lands on the floor, on a plinth, or inside the bay. The jug is carried over the bay, the mop rest, or the sign. Or the jug is not stood back upright on its spot.
- **SOP rule broken:** Step 1. Keep the lip inside the rim, pour in one go to the fill line, then stand the jug upright back on the jug spot.
- **Coaching note:** lip inside the rim, pour to the line, jug straight back. Water on the floor is not part of the task.

Violation: Bucket not held, or held wrong

- **Visible cue:** the **left gripper** is off the left side lug during the pour or during a wring. The bucket skates, turns, or tips. The **left gripper** lifts the bucket, drags it, or closes on the rim or the wringer instead of the lug.
- **SOP rule broken:** Steps 1, 2.2, and 2.3. The **left gripper** closes on the left side lug and holds the bucket still on the floor for the whole pour and for every wring.
- **Coaching note:** the left lug is the only hold. Close on it before the jug or the mop moves, and hold until they are clear.

Violation: Mop taken up or held wrong

- **Visible cue:** the **right gripper** closes on the head, the pad, or the handle away from the grip band. The handle turns or slips in the grip. The head is swung over the bucket, the cabinets, or the bay while it is up. Or the arm extends to follow the head instead of the handle doing the reaching.
- **SOP rule broken:** Steps 2.1 to 4.2. Close on the grip band only, keep the handle upright when the head is up, and let the handle do the reaching.
- **Coaching note:** grip band only, and let the pole reach for you. If the arm is stretching, the hold is too low.

Violation: Dip wrong or skipped

- **Visible cue:** the head goes into the bay without being dipped. It is moved up and down fewer or more than twice. It is stirred round the bucket. It is knocked against the sides. Or it is dipped into the wringer instead of the water.
- **SOP rule broken:** Step 2.2. Lower the head into the water clear of the wringer and move it up and down twice.
- **Coaching note:** straight down, twice up and down, straight up. The dip wets the pad. It does not wash the bucket.

Violation: Wring skipped or wrong

- **Visible cue:** the head goes to the floor without being set in the wringer. The handle is pressed fewer or more than twice. The handle is levered sideways in the wringer. Water still runs off the head as it leaves the bucket. Or the second dip and wring before the edge pass is skipped.
- **SOP rule broken:** Steps 2.3 and 4.1. Set the head in the wringer, press the handle straight down twice, and hold it clear for 2 seconds with no water running off, before the laps and again before the edge pass.
- **Coaching note:** two presses, then watch it for two seconds. A dripping head leaves puddles, not a clean floor.

Violation: First lap not set right

- **Visible cue:** the head enters the bay somewhere other than the front-left corner. It starts against the left plinth instead of just clear of it. It starts with a corner lifted. Or its long side is not across the bay.
- **SOP rule broken:** Step 3.1. Set the head flat at the front-left corner, long side across the bay, just clear of the left plinth.
- **Coaching note:** set it, look at it, then push. A pattern that starts wrong cannot finish right.

Violation: Lap run wrong

- **Visible cue:** the head stops mid-lap, jerks or changes speed, turns as it runs, is dragged sideways during the push, or a corner of the pad rides up off the floor.
- **SOP rule broken:** Step 3.2. Run each lap in one steady push with the pad flat the whole way.
- **Coaching note:** one steady push, all the way. Stopping mid-lap leaves a mark you cannot push out.

Violation: Head run onto the kick rail or a plinth

- **Visible cue:** the head rides up onto the kick rail or onto a plinth. Or water is pushed up onto a board and left there.
- **SOP rule broken:** Steps 3.2, 3.3, and 4.2. Stop the head just short of the kick rail, keep it just clear of each plinth on the laps, and keep it against the line but off the board on the edge pass.
- **Coaching note:** the floor is the whole run. The boards are not part of it.

Violation: Laps not overlapped

- **Visible cue:** a dry band is left between two wet strips. A lap is set clear of the strip before it. Or a lap is set so far onto the wet strip that it covers almost no dry floor.
- **SOP rule broken:** Step 3.3. Set each new lap so about a third of the head sits on the wet strip the last lap left.
- **Coaching note:** a third on the wet, the rest on the dry. Read the wet line before you set the head.

Violation: Laps worked in the wrong direction, or a lap left out

- **Visible cue:** the step overs run right to left. The laps run out of order. Fewer or more than three laps are run. Or the floor at a plinth is never covered by a lap.
- **SOP rule broken:** Steps 3.1 to 3.3. Run three laps from the left of the bay rightward: lap 1, step over, lap 2, step over, lap 3.
- **Coaching note:** left side first, work right, finish on the right. Three laps, two step overs, one shape.

Violation: Edge pass broken up or run out of order

- **Visible cue:** the head leaves the floor between edges. The edge pass is run as separate strokes instead of one unbroken run. The edges are run in an order other than front, left, back, right. Or an edge is skipped.
- **SOP rule broken:** Step 4.2. Run the head round the bay in one unbroken move: front edge, left edge, back edge, right edge.
- **Coaching note:** one lap round the outside, in one go. Lifting between edges is how an edge gets missed.

Violation: Edge not run tight to the line

- **Visible cue:** a dry band is left between the wet floor and the kick rail or a plinth. The head runs along an edge with its long side turned away from the line. Or a corner of the bay is cut instead of turned into.
- **SOP rule broken:** Step 4.2. Keep the long side of the head tight against the edge it is on, and turn into each corner.
- **Coaching note:** the head follows the line, not near it. The corners are turns, not shortcuts.

Violation: Head turned by twisting the wrist

- **Visible cue:** the **right gripper** twists at the wrist to turn the head at a corner. Or the wrist reaches a limit and the head stalls or skids instead of turning.
- **SOP rule broken:** Step 4.2. Turn the head by moving the handle round it, not by twisting the wrist.
- **Coaching note:** walk the handle round the corner. The wrist has less turn in it than the corner needs.

Violation: Mop rested wrong

- **Visible cue:** the mop is laid on the floor of the bay, leaned on a plinth or a cabinet, stood with the head inside the bay, or left leaning, slipping, or fallen out of its cradle. Or the head is dropped rather than set down.
- **SOP rule broken:** Step 5. Stand the handle in the cradle with the head flat on the floor outside the bay, and let go only once it stands.
- **Coaching note:** into the cradle, head down outside the bay, then let go. Watch it stand before the arm leaves.

Violation: Sign opened or staged wrong

- **Visible cue:** the sign is lifted off the floor, slid along the floor, or carried. The **right gripper** touches it. The bottom panel slides or lifts. The sign folds back, tips, or stands on one set of feet. The printed face points into the bay instead of out to the front. Or any part of it ends up inside the bay.
- **SOP rule broken:** Step 6. Turn the top panel up and over toward the front with the **left gripper**, and leave the sign standing open at the sign place, printed face out.

- **Coaching note:** turn the top panel over and let go only once it stands on its own. The sign never leaves the floor.

Violation: Required check not followed

- **Visible cue:** a check named in a step is skipped. Or a check is made and the fault it finds is left uncorrected: water short of the fill line, a dripping head, a dry band between laps, a dry band along an edge, a mop that will not stand, or a sign that folds back.
- **SOP rule broken:** Steps 1, 2.3, 3.1, 3.2, 3.3, 4.1, 4.2, 5, and 6. Run each check and correct what it finds by the retry written in that step.
- **Coaching note:** a check is not done until what it found has been put right.

Violation: Dropped, knocked over, or spilled

- **Visible cue:** the mop, the jug, or the sign is dropped or knocked over. The bucket is knocked off its place or tipped. Water is slopped out of the bucket onto the floor. Or an arm sweeps a tool out of place on its way in or out.
- **SOP rule broken:** Steps 1 to 6. Nothing is dropped, knocked out of its place, or spilled, and every arm comes in and goes out clear of the bucket, the mop rest, and the sign.
- **Coaching note:** check the path and the landing place before the arm moves, and set things down before letting go.

Violation: Wrong arm used

- **Visible cue:** the **left gripper** touches the mop, the jug, the wringer, or the floor of the bay. The **left gripper** enters the bay during Step 3 or Step 4. The **right gripper** holds the bucket's left side lug or touches the sign. Or an object is handed from one gripper to the other.
- **SOP rule broken:** Steps 1 to 6. The **right gripper** pours, mops, and rests the mop. The **left gripper** holds the bucket lug and opens the sign. Nothing changes hands.
- **Coaching note:** right gripper works the tools, left gripper holds the bucket and opens the sign. The left arm never enters the bay.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a dry band, a scuff, or a streak on the floor, an edge not run tight, water left on a board or outside the bay, the mop out of its cradle, the jug off its spot, the sign folded or facing the wrong way, an arm short of home, or a gripper not fully open.
- **SOP rule broken:** Step 7. Confirm the floor, the edges, the furniture, the tools, the sign, and the base. Then return both arms home with grippers open and stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Camera cannot read the floor:** glare on the floor, or a shadow from the arms, hides part of the bay, so whether it is clear cannot be judged.
- **Camera cannot read the floor-to-board line**, so whether the edge pass ran tight cannot be judged.
- **Camera cannot read the fill line** inside the bucket.
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Base fault:** a brake or lock that will not hold, or a base that rolls, creeps, or turns with the lock set.
- **Furniture not fixed:** a kick rail, plinth, or cabinet that rocks or shifts under a correct push.
- **Bay too large for one park:** a bay a correctly parked base cannot cover, so the kick rail or a plinth cannot be reached without extending an arm.
- **Defective mop:** a head that will not lie flat, a hinge that spins loose, a handle that turns in the grip whatever the hold, or a pad that is torn or shedding onto the floor.
- **Defective pad:** one that goes dry before the edge pass is finished, or that leaves a streak of its own on a correctly wrung head.
- **Defective bucket or wringer:** a bucket that tips under a correct wring, a left side lug that will not hold in one gripper's opening, or a wringer that will not squeeze the pad damp in two presses.
- **Defective jug:** one that leaks, that tips when gripped, whose handle will not sit inside one gripper's opening, or that holds too little water to reach the fill line in one pour.
- **Defective sign:** one that folds back on its own however correctly it is opened, that will not stand on its own feet, whose grab hole will not hold in one gripper's opening, or whose bottom panel slides on the floor when the top panel is turned up.
- **Defective floor:** a chip, a stain in the floor itself, or a worn patch that a correct lap and a correct edge pass will not lift. Or a loose tile that rocks under the head.
- **Debris in the bay:** grit, crumbs, or loose rubbish on the floor at episode start, which a mop pushes into a pile instead of lifting.
- **A place turns out to sit outside its arm's comfortable reach** with the base correctly parked, so the jug spot, the bucket, the wringer, the mop rest, the sign, or a corner of the bay cannot be reached without extending or folding the arm.

Annotation subtasks (from SOP)

1. Hold the bucket by its left side lug
2. Take the water jug from the jug spot
3. Pour the jug into the bucket to the fill line
4. Return the water jug to the jug spot
5. Take the mop from the mop rest
6. Dip the mop head in the bucket
7. Wring the mop head in the wringer
8. Bring the mop head into the bay
9. Run one S lap across the bay
10. Step the head over to the next lap
11. Run the edge pass round the bay
12. Return the mop to the mop rest
13. Turn the sign's top panel up and open
14. Return both arms home and end the episode